

Oil Price Shocks and India's Terms of Trade

An Empirical Assessment

Rehan Chouhan · Gaurav Arora · Munna Meetha · Sanjana

October 2026

International Economics · Group project

March 2026: India's oil shock, again

+46%

Brent crude, Feb to Mar 2026 (\$71 to \$104 a barrel)

\$113

India's crude basket in March, up from \$69 in February

Rs 94.65

per US dollar at end-March 2026, the weakest year-end rate on record

—24⁰%

India's merchandise terms of trade after the last shock, FY2020-21 to FY2022-23

Same pattern in 1973, 1979, 1990 (just before the 1991 BoP crisis) and 2022.

How much do oil shocks move India's terms of trade, and does it matter why oil became expensive?

Question and hypotheses

What we measure

Net barter terms of trade (L21)

$$\frac{\text{export price index}}{\text{import price index}} \times 100$$

Higher means each unit of exports buys more imports.

India imports about 90% of the crude it refines (RBI Table 32) and is the world's 3rd-largest oil consumer (5.4% of demand, EIA 2024).

H1 Oil price rises worsen India's terms of trade

Offer curves (L22): the oil exporter restricts supply, India's ToT ray flattens

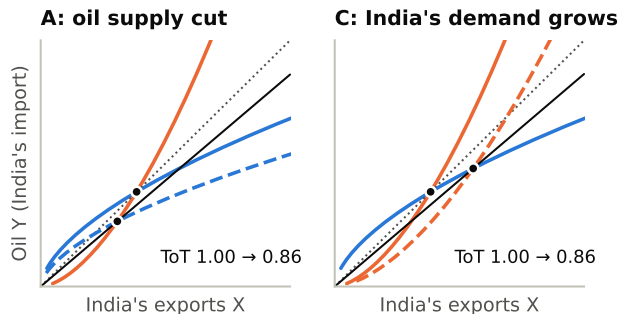
H2 Rises and falls have unequal effects

OPEC+ market power (L16) defends price floors; Mork (1989) asymmetry

H3 Supply-driven rises hurt more than demand-driven

Offer curves, case B: a world boom also lifts demand for India's exports

Theory in action: offer curves for India



India's ToT is the slope of the ray through the equilibrium (P_X/P_Y). A flatter ray means each unit of India's exports buys less oil. Data: after the 2022 shock, India's merchandise ToT fell 24% (FY2020-21 to FY2022-23).

A supply cut works like a tariff

OPEC+ offers less oil per unit of India's exports: its ToT improves, India's falls (L22).

India is not a price-taker

As the 3rd-largest buyer, India's own demand growth also worsens its ToT (L11).

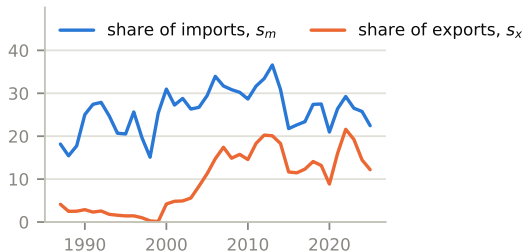
World booms shift both curves

India's export demand rises too, so the sign is an empirical question: H3.

How big should the hit be?

$$d \ln ToT \approx (s_x - s_m) \cdot d \ln P_{oil} \quad \text{oil's share in exports minus its share in imports}$$

Oil in India's merchandise trade (%)



−0.25 → −0.10

Benchmark $s_x - s_m$, FY1999-00 to FY2025-26: net oil exposure more than halved

0.06 → 4.37

Revealed comparative advantage in refined petroleum, 1999 to 2025 (Balassa, LO8)

0.55 vs 0.10

Grubel-Lloyd for oil, aggregated vs crude and products separately (L17, L19)

What we know, and the gap

Why ToT matter

Harberger (1950);
Laursen & Metzler
(1950): a worse ToT
lowers real income and
the trade balance.

How much is debated:
Mendoza (1995) vs
Schmitt-Grohé & Uribe
(2018).

Oil drives ToT

Backus & Crucini
(2000): oil explains
much of the variation
in ToT.

Kilian, Rebucci &
Spatafora (2009): the
effect depends on the
kind of oil shock.

Shocks differ

Hamilton (1983, 2003);
Mork (1989):
asymmetry.

Kilian (2009);
Baumeister & Hamilton
(2019); Känzig (2021):
supply vs demand.

India

Bhanumurthy, Das &
Bose (2012); Ghosh
(2011); Tiwari & Olayeni
(2013); Deheri & Sahu
(2024).

Focus: trade balance,
inflation, the rupee.

Gap. Indian studies model the trade balance, inflation or the rupee. We found none that models India's terms of trade directly, with a theory benchmark, identified supply and demand shocks, and the 2022 and 2026 episodes.

Data: official sources, checked line by line

Official agencies and published replication data only

RBI Handbook 2025-26	UNCTADstat
DGCI&S trade indices	UN Comtrade
PPAC (Indian basket)	US EIA BIS
World Bank Pink Sheet	Baumeister-Hamilton
World Bank WDI	Känzig (AER data)

Monthly: Apr 2019–Jun 2026 (87 months)

Annual: FY1970-71 to FY2025-26

Merchandise ToT: FY1994-95 to FY2025-26

o failed checks

recomputed indices, cross-source comparisons,
fiscal-year alignment

Problems we found in official tables

- RBI's base-year ToT series contradict each other: +21.7% vs -21.0% (1999–2007)
- RBI Table 32 repeats the 2000–07 rows for 1990–97
- DGCI&S export volume index jumps between 54 and 2001, so we use price indices only
- Three published RBI ToT values disagree with RBI's own indices

Method in four steps

1

Unit-root tests

ADF, Phillips–Perron,
KPSS, Zivot–Andrews

All series stationary
in differences, none
 $I(2)$, so ARDL is valid

2

ARDL bounds test

H1: short-run vs
long-run effect,
speed of adjustment

Works with mixed
 $I(0)/I(1)$ and 54 years;
short vs long run as
in L13

3

Nonlinear ARDL

H2: separate effects
of oil price rises and
falls

Wald tests of
long-run and
short-run symmetry

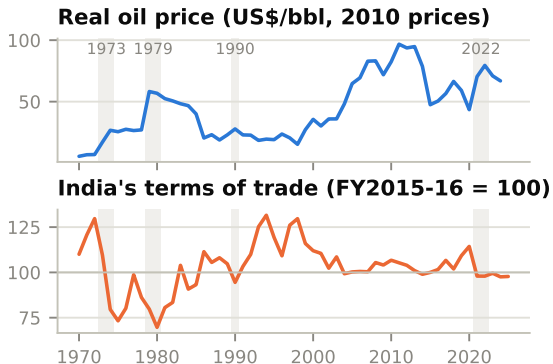
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Shock split + local projections

H3: supply- vs
demand-driven oil
prices; monthly
timing

Baumeister–
Hamilton and Känzig
shocks are not
caused by India

H1: oil hurts India's terms of trade, and fast



—0.24

Short-run elasticity: a 10% oil rise cuts ToT by 2.4% the same year ($p < 0.001$, robust s.e.)

31% a year

of any gap closes each year (error correction $\alpha = -0.31$): the hit is large but temporary

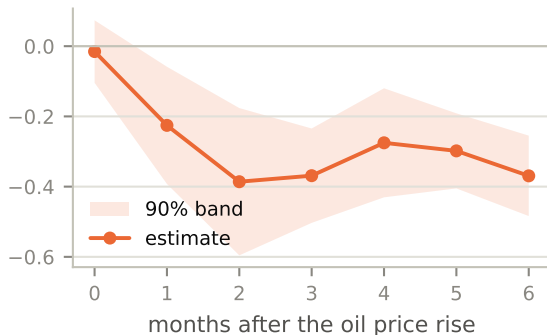
—0.09

Long-run elasticity, not significant; bounds $F = 4.74$ ($p = 0.07$)

—0.19 to —0.28 in six specifications; diagnostics pass; CUSUM stable.

The hit arrives within two to three months

ToT response (%) to a 1% Brent rise



—0.39

Elasticity two months after an oil rise; almost nothing in month 0 (monthly data, Apr 2019–Jun 2026)

1.70

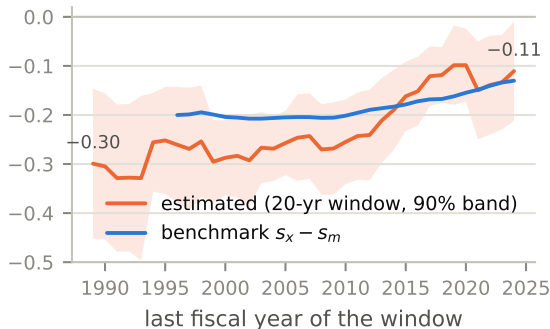
Benchmark test: theory predicts a coefficient of 1 on $(s_x - s_m) \Delta \ln P_{oil}$; 1 is not rejected at 5% ($p = 0.07$)

+18% vs +16%

2026 so far: export prices rose as fast as import prices (Feb–Apr), so ToT has not clearly fallen yet

Symmetric, and half as large as before

Oil elasticity of India's ToT



H2 not supported

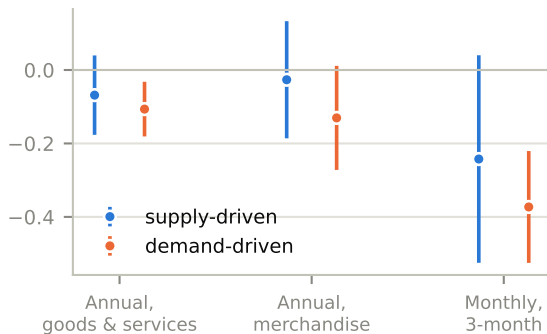
Short run: rise -0.27 , fall -0.22 ; symmetry never rejected ($p = 0.54-0.91$). A long-run asymmetry appears only with the 1970s in the sample ($p = 0.28$ from 1980).

Exposure halved

Elasticity -0.30 (window to 1989) to -0.11 (to 2024). It tracks the benchmark ($-0.25 \rightarrow -0.10$) as refined-fuel exports grew: the refining advantage works as a hedge ($RCA \approx 4$).

H3: how much oil rises matters more than why

ToT effect of a 1% oil rise, by cause



H3 not supported

Demand-driven rises hurt at least as much as supply-driven ones ($p = 0.53$ – 0.71 for a difference).

Why

A world boom also lifts the prices of gold, coal, fertiliser, edible oil and metals that India imports.

Cross-check

Känzig's OPEC-news shocks (about +10% on oil): India's ToT falls 4–5% two to three months later.

What it means for India

What we found

- H1** A 10% oil rise cuts India's ToT by 2.4% within a year, mostly within 2–3 months
- H2** Rises and falls matter about equally
- H3** Demand-driven rises hurt as much as supply-driven ones
 - + Exposure has halved since the 1990s: refining is India's hedge

What India can do

Buffers for both directions: strategic reserves and a fuel-tax buffer that cuts duties in spikes and rebuilds them in slumps

Grow the refining hedge: more refined and petrochemical exports raise s_x (L08, L16)

Diversify suppliers: more sources weaken any one seller's market power (L16)

Cut commodity dependence: energy transition and ethanol blending, since demand booms hurt too

Thank you

Questions welcome

Backup: oil episodes, fiscal-year averages

Episode	Fiscal years	Real oil	ToT (G&S)	ToT (merch.)
1973 OPEC embargo	1972-73 to 1974-75	+284.1%	−38.6%	−
1979 Iranian revolution	1978-79 to 1980-81	+110.2%	−19.2%	−
1990 Gulf war	1989-90 to 1990-91	+21.1%	−9.8%	−
2008 price spike	2006-07 to 2008-09	+19.9%	+4.9%	+5.8%
2014-16 price collapse	2013-14 to 2015-16	−49.9%	−1.0%	+19.4%
2020 COVID crash	2019-20 to 2020-21	−26.4%	+4.7%	+9.6%
2022 Russia-Ukraine	2020-21 to 2022-23	+82.8%	−14.4%	−23.7%

2008 looks mild in annual averages because the spike lasted only months; the 2014–16 collapse shows the merchandise gain from cheap oil.

Backup: unit-root tests (p-values)

Series	ADF lvl	ADF diff	PP lvl	PP diff	KPSS lvl	ZA lvl	ZA break
ln ToT goods & services	0.057	0.000	0.050	0.000	0.100	0.391	1985
ln ToT merchandise (FY1994-)	0.271	0.001	0.392	0.000	0.010	0.062	2011
ln real oil price	0.053	0.000	0.050	0.000	0.013	0.533	1999
ln real non-energy prices	0.355	0.000	0.395	0.000	0.100	0.911	2005
ln real gold price	0.945	0.000	0.582	0.000	0.010	0.942	2006

ADF and PP: null of a unit root. KPSS: null of stationarity. ZA: unit root with one break. Every series is stationary in differences, so none is $I(2)$.

Backup: ARDL main estimates

	Estimate	Std. error	p-value
Short-run oil elasticity (same year)	−0.241	0.049	0.000
Long-run oil elasticity	−0.092	0.059	0.119
Long-run non-oil commodity elasticity	−0.249	0.177	0.159
Error-correction speed α	−0.308	0.086	0.001

Bounds $F = 4.74$, $p = 0.025$ against $I(0)$ and 0.070 against $I(1)$. Diagnostics (p): Breusch–Godfrey 0.90 , Breusch–Pagan 0.019 (so HC1 robust s.e.: 0.040 for the short-run effect), Jarque–Bera 0.25 , RESET 0.84 ; CUSUM and CUSUMSQ inside 5% bands.

Backup: ARDL robustness

Specification	N	SR oil	t	LR oil	ECT α	F	p (I(1))
Baseline: oil + non-oil (G&S ToT)	54	−0.241	−4.931	−0.092	−0.308	4.743	0.070
Oil only	54	−0.195	−4.577	−0.084	−0.241	4.064	0.173
Oil + non-oil + gold	54	−0.217	−4.738	−0.061	−0.310	3.538	0.149
Brent instead of average crude	54	−0.236	−4.972	−0.089	−0.307	4.709	0.072
Post-1980 sample	44	−0.211	−4.965	−0.192	−0.355	3.651	0.172
Merchandise ToT, FY1994-2024	30	−0.277	−3.457	−0.146	−0.368	2.084	0.506

Backup: NARDL symmetry tests

Specification	N	L^+	L^-	p (LR)	π^+	π^-	p (SR)
Baseline FY1970-2024, with non-oil control	54	-0.162	-0.228	0.035	-0.271	-0.215	0.620
No control	54	-0.199	-0.285	0.017	-0.233	-0.192	0.735
Brent	54	-0.157	-0.221	0.036	-0.261	-0.217	0.693
Extra lag of ΔToT	53	-0.119	-0.180	0.051	-0.239	-0.215	0.831
FY1975-2024	49	-0.106	-0.161	0.077	-0.217	-0.204	0.913
FY1980-2024 (drops the 1970s shocks)	44	-0.114	-0.145	0.275	-0.245	-0.172	0.537
Merchandise ToT FY1994-2024	30	-0.187	-0.075	0.131	-0.234	-0.308	0.746

$L^\pm$: long-run effects of rises and falls; $\pi^\pm$: short-run. Short-run symmetry is never rejected; the long-run asymmetry needs the 1970s.

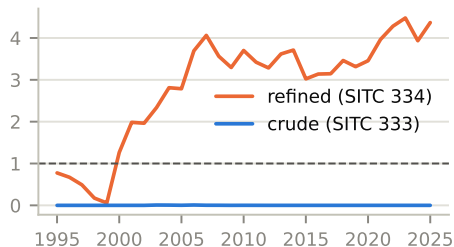
Backup: supply- vs demand-driven oil prices

Sample	N	Supply	s.e.	Demand	s.e.	p (equal)
Annual G&S ToT, FY1975-2025	51	−0.068	0.066	−0.106	0.045	0.707
Annual merchandise ToT, FY1995-2024	31	−0.027	0.097	−0.130	0.086	0.547
Monthly merchandise ToT, 3-month cumulative, 2019-26	82	−0.242	0.172	−0.373	0.093	0.535

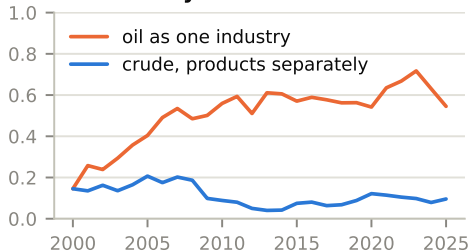
Supply part = fitted contribution of Baumeister–Hamilton supply shocks to the oil price change;
demand part = activity, consumption and inventory shocks. Känzig news shock: annual −0.95% per shock (s.e. 0.49); monthly −4.2% and −4.7% at 2 and 3 months.

Backup: the refining hedge in trade-theory terms

Revealed comparative advantage



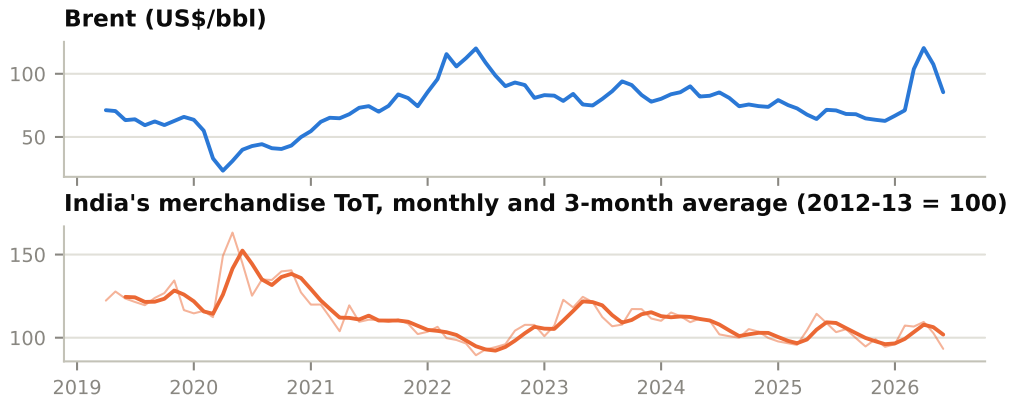
Grubel-Lloyd index of oil trade



Refined exports are worth 1.13–1.41 times crude imports per tonne (2000–2025), outside the $\pm 15\%$ band that L2O uses for horizontal trade in every year except 2011 and 2012. So this is vertical, value-adding trade: India imports crude and sells refined fuel.

Year	Crude import (US\$/t)	Refined export (US\$/t)	Ratio
2005	361.40	462.95	1.28
2015	368.81	520.32	1.41
2022	713.15	996.29	1.40
2024	585.45	721.25	1.23
2025	528.03	672.85	1.27

Backup: monthly terms of trade, 2019–2026



Backup: limitations and the data fixes we made

- Unit value indices are not pure price indices: they mix composition and quality changes
- Monthly DGCI&S data start in April 2019; earlier monthly series are not published online
- DGCI&S's two monthly bases (2012-13 and 2022-23) disagree month to month, so no episode is judged on a single month
- The goods-and-services ToT includes services (software), which dilutes the oil effect
- Imports are valued CIF and exports FOB, so freight and insurance spikes also move measured ToT (L14)
- Fixes: RBI base-year series linked at overlaps; NTT recomputed from the price indices; duplicated RBI rows dropped; WDI 1960s deflators excluded; mis-served PPAC file set aside
- The 2026 Hormuz shock is still unfolding (ToT data to June 2026)